Appendix B: Research Gap Analysis and Publication Strategy

B.1 Research Gap Analysis

B.1.1 Gap Identification Methodology

Research gaps are identified through systematic analysis of 240 papers (30 per research direction) using the following criteria:

1. Evidence-supported gaps: Gaps consistently observed across multiple papers within a direction
2. Scientific significance: Gaps representing fundamental challenges rather than incremental improvements
3. Novelty potential: Gaps offering opportunities for novel contributions to the field
4. Experimental verifiability: Gaps addressable through rigorous experimental validation
5. Publication potential: Gaps aligned with high-impact research questions in robotics

Gaps are distinguished from speculative opportunities by requiring:

- Explicit evidence: At least 5 papers within a direction demonstrating the limitation
- Quantifiable metrics: Clear measurement criteria for gap closure
- Feasibility assessment: Realistic experimental approaches for validation

B.1.2 RVI-01: Reproducible Benchmarking — Gap Analysis

Gap 1.1: Inspection-Specific Benchmarking Protocols

Evidence: 28 of 30 reviewed papers evaluate manipulation, navigation, or domain-specific tasks (autonomous driving, agriculture); only 2 papers address visual inspection 1 , 2

Scientific significance: High — Inspection requires different evaluation criteria than manipulation (documentation quality vs. task success)

Novelty potential: Very high — No existing benchmark systematically evaluates autonomous visual inspection

Feasibility: High — Can leverage existing datasets (YCB, ShapeNet) and simulation environments (Isaac Sim, Gazebo)

Experimental approach:

- Define standardized inspection tasks (object documentation, surface coverage, defect localization)
- Establish evaluation metrics (coverage completeness, acquisition quality, repeatability)
- Implement controlled environmental variations (lighting, object placement, occlusion)
- Validate across multiple robot platforms (manipulator, mobile robot, humanoid)

Publication potential: Very high — Benchmarking papers attract broad interest and citations

Gap 1.2: Humanoid Platform Validation

Evidence: Only 2 of 30 papers evaluate humanoid robots 1 , 22 ; 18 papers evaluate manipulators, 8 papers evaluate wheeled robots, 2 papers evaluate UAVs

Scientific significance: High — Humanoid platforms introduce unique challenges (bipedal

stability, whole-body kinematics) not addressed by manipulator studies

Novelty potential: High — Humanoid inspection is underexplored

Feasibility: Medium — Requires humanoid platform access (Unitree G1, Boston Dynamics Atlas)

Experimental approach:

- Compare humanoid, manipulator, and wheeled robot performance on standardized inspection tasks
- Quantify trade-offs (workspace coverage, execution time, repeatability)
- Validate kinematic feasibility and stability constraints

Publication potential: High — Humanoid robotics is a high-impact research area

Gap 1.3: Repeatability Metrics

Evidence: 26 of 30 papers evaluate robustness to perturbations; only 4 papers address repeatability 1 , 2 , 13 , 18

Scientific significance: High — Repeatability is critical for measurement applications but underexplored in benchmarking

Novelty potential: Medium — Repeatability is studied in metrology but not robotics benchmarking

Feasibility: Medium — Requires independent measurement systems (laser trackers, photogrammetry)

Experimental approach:

- Define repeatability metrics (pose consistency, image attribute consistency, measurement condition consistency)
- Conduct repeated trials across sessions and environmental variations
- Validate against independent ground truth measurements

Publication potential: Medium — Repeatability is important but less central to robotics research

Gap ranking: 1.1 > 1.2 > 1.3 (based on novelty, feasibility, publication potential)

B.1.3 RVI-02: Target Identification and Localization — Gap Analysis

Gap 2.1: Multi-Instance Identification in Clutter

Evidence: 22 of 30 papers address single-object detection or pose estimation; only 8 papers address multi-object scenarios 6 , 10 , 20 , 24

Scientific significance: Medium — Multi-instance identification is challenging but not fundamentally novel

Novelty potential: Low — Object detection is well-studied; incremental improvements are harder to publish

Feasibility: High — Can leverage existing deep learning methods (YOLO, Mask R-CNN)

Experimental approach:

- Evaluate existing methods on cluttered multi-object inspection scenarios
- Develop instance-specific identification strategies (e.g., using unique features or spatial context)
- Validate on standardized datasets (YCB, COCO)

Publication potential: Medium — Incremental improvements in object detection are harder to publish in top venues

Gap 2.2: Geometric Measurement Accuracy

Evidence: 25 of 30 papers emphasize pose estimation for manipulation; only 5 papers address measurement accuracy 14 , 22 , 28 , 29 , 30

Scientific significance: Medium — Measurement accuracy is important for inspection but not fundamentally novel

Novelty potential: Low — Pose estimation accuracy is well-studied

Feasibility: High — Can leverage existing pose estimation methods with calibration

Experimental approach:

- Evaluate pose estimation accuracy against independent ground truth (laser tracker, photogrammetry)
- Quantify measurement uncertainty and calibration errors
- Validate on standardized objects (YCB)

Publication potential: Low — Incremental improvements in pose estimation accuracy are harder to publish

Gap 2.3: Humanoid Integration

Evidence: 28 of 30 papers do not consider humanoid kinematics; only 2 papers address humanoid-specific challenges 1 , 22

Scientific significance: Medium — Humanoid integration is important but not fundamentally novel for object detection

Novelty potential: Medium — Humanoid-specific object detection is underexplored

Feasibility: Medium — Requires humanoid platform access

Experimental approach:

- Evaluate object detection and pose estimation on humanoid platforms
- Quantify effects of whole-body motion and camera placement
- Validate on standardized tasks

Publication potential: Medium — Humanoid robotics is high-impact, but object detection is well-studied

Gap ranking: 2.1 > 2.3 > 2.2 (based on novelty, feasibility, publication potential)

B.1.4 RVI-03: Viewpoint and Motion Planning — Gap Analysis

Gap 3.1: Humanoid-Specific Planning

Evidence: 26 of 30 papers target manipulators or UAVs; only 4 papers address humanoid-specific challenges 1 , 4 , 14 , 22

Scientific significance: Very high — Humanoid planning requires whole-body kinematics and bipedal stability constraints not addressed by manipulator methods

Novelty potential: Very high — Humanoid-specific inspection planning is underexplored

Feasibility: Medium — Requires humanoid platform access and kinematic modeling

Experimental approach:

- Develop humanoid-specific viewpoint planning incorporating whole-body kinematics and stability constraints
- Compare against adapted manipulator methods
- Validate on physical humanoid platform (Unitree G1)

Publication potential: Very high — Planning research is central to robotics; humanoid-specific methods offer novelty

Gap 3.2: Physical Execution Validation

Evidence: 18 of 30 papers rely primarily on simulation; only 12 papers include physical robot experiments 1 , 2 , 3 , 4 , 5 , 9 , 13 , 18 , 24 , 25 , 29 , 30

Scientific significance: High — Simulation cannot validate kinematic feasibility, collision avoidance, or execution reliability

Novelty potential: Medium — Physical validation is important but not fundamentally novel

Feasibility: Medium — Requires physical robot platform and controlled experimental setup

Experimental approach:

- Validate planned trajectories on physical robot platform
- Quantify execution success rate, kinematic feasibility, collision avoidance
- Compare simulation predictions against physical measurements

Publication potential: High — Physical validation strengthens publication impact

Gap 3.3: Quality Integration

Evidence: 27 of 30 papers optimize coverage without integrating image quality metrics; only 3 papers address quality-aware planning 1 , 12 , 13

Scientific significance: High — Quality-aware planning integrates perception, planning, and control in a closed loop

Novelty potential: High — Integrated quality-aware planning is rare

Feasibility: High — Can build on existing exposure control and quality assessment methods

Experimental approach:

- Develop quality-aware viewpoint planning integrating image quality assessment
- Compare against coverage-first planning with post-hoc quality filtering
- Validate on physical robot platform

Publication potential: Very high — Quality-aware planning is novel and relevant to robotics and computer vision

Gap ranking: 3.1 > 3.3 > 3.2 (based on novelty, feasibility, publication potential)

B.1.5 RVI-04: Active Visual Coverage — Gap Analysis

Gap 4.1: Quality-Aware NBV Planning

Evidence: 27 of 30 papers optimize coverage or information gain without integrating image quality; only 3 papers address quality-aware planning 1 , 12 , 13

Scientific significance: High — Quality-aware NBV planning integrates perception, planning, and control

Novelty potential: High — Integrated quality-aware NBV planning is rare

Feasibility: High — Can build on existing NBV methods and quality assessment

Experimental approach:

- Develop quality-aware NBV planning integrating image quality assessment into viewpoint selection
- Compare against coverage-first NBV with post-hoc quality filtering
- Validate on physical robot platform

Publication potential: High — Quality-aware planning is novel and relevant to robotics

Gap 4.2: Humanoid Platform Evaluation

Evidence: 27 of 30 papers evaluate manipulators or UAVs; only 3 papers address humanoid-specific challenges 14 , 22 , 25

Scientific significance: High — Humanoid NBV planning requires whole-body kinematics and stability constraints

Novelty potential: High — Humanoid-specific NBV planning is underexplored

Feasibility: Medium — Requires humanoid platform access

Experimental approach:

- Develop humanoid-specific NBV planning
- Compare against adapted manipulator methods
- Validate on physical humanoid platform

Publication potential: High — Humanoid robotics is high-impact

Gap 4.3: Sim-to-Real Validation

Evidence: 18 of 30 papers rely primarily on simulation; only 12 papers include physical robot experiments 1 , 3 , 4 , 5 , 9 , 13 , 15 , 20 , 21 , 24 , 25 , 29

Scientific significance: Medium — Sim-to-real validation is important but not fundamentally novel

Novelty potential: Low — Sim-to-real transfer is well-studied

Feasibility: Medium — Requires physical robot platform

Experimental approach:

- Validate simulated NBV planning on physical robot platform
- Quantify sim-to-real gap in coverage, execution time, success rate
- Compare simulation predictions against physical measurements

Publication potential: Medium — Sim-to-real validation strengthens publication impact

Gap ranking: 4.1 > 4.2 > 4.3 (based on novelty, feasibility, publication potential)

B.1.6 RVI-05: Repeatable Visual Acquisition — Gap Analysis

Gap 5.1: Acquisition Geometry Repeatability

Evidence: 26 of 30 papers measure robot pose repeatability; only 4 papers address acquisition geometry or image attribute consistency 12 , 24 , 25 , 26

Scientific significance: High — Acquisition geometry repeatability extends beyond robot pose to encompass camera observation geometry and measurement conditions

Novelty potential: Medium — Repeatability is studied in metrology but not robotics

Feasibility: Low — Requires precise calibration and independent measurement systems

Experimental approach:

- Define acquisition geometry repeatability metrics (camera pose, framing, resolution, illumination)
- Conduct repeated trials across sessions and environmental variations
- Validate against independent ground truth measurements

Publication potential: Medium — Repeatability is important but less central to robotics research

Gap 5.2: Quality Metric Integration

Evidence: 28 of 30 papers do not integrate image quality metrics into repeatability evaluation; only 2 papers address quality consistency 12 , 25

Scientific significance: Medium — Quality metric integration is important for inspection but not fundamentally novel

Novelty potential: Low — Quality assessment is well-studied

Feasibility: Medium — Can build on existing quality assessment methods

Experimental approach:

- Integrate image quality metrics (blur, exposure, occlusion) into repeatability evaluation
- Quantify quality consistency across repeated trials
- Validate on physical robot platform

Publication potential: Low — Incremental integration of quality metrics is harder to publish

Gap 5.3: Humanoid Platform Evaluation

Evidence: 29 of 30 papers do not evaluate humanoid robots; only 1 paper addresses humanoid-specific challenges 24

Scientific significance: Medium — Humanoid repeatability is important but not fundamentally novel

Novelty potential: Medium — Humanoid-specific repeatability is underexplored

Feasibility: Medium — Requires humanoid platform access

Experimental approach:

- Evaluate acquisition repeatability on humanoid platform
- Quantify effects of whole-body motion and bipedal stability
- Validate on physical humanoid platform

Publication potential: Medium — Humanoid robotics is high-impact, but repeatability is less central

Gap ranking: 5.1 > 5.3 > 5.2 (based on novelty, feasibility, publication potential)

B.1.7 RVI-06: Acquisition Quality Control — Gap Analysis

Gap 6.1: Integrated Quality-Aware Planning

Evidence: 27 of 30 papers address exposure control or quality assessment in isolation; only 3 papers integrate quality into planning 10 , 12 , 13

Scientific significance: Very high — Integrated quality-aware planning closes the loop between perception, planning, and control

Novelty potential: Very high — Integrated quality-aware planning is rare

Feasibility: High — Can build on existing exposure control and quality assessment methods

Experimental approach:

- Develop integrated quality-aware planning combining exposure control, quality assessment, and viewpoint selection
- Compare against reactive quality control (post-hoc filtering and re-observation)
- Validate on physical robot platform

Publication potential: Very high — Quality-aware planning is novel and relevant to robotics and computer vision

Gap 6.2: Inspection-Specific Quality Metrics

Evidence: 25 of 30 papers use generic quality metrics (gradient, entropy, noise); only 5 papers address inspection-specific attributes 8 , 12 , 14 , 15 , 28

Scientific significance: Medium — Inspection-specific metrics are important but not fundamentally novel

Novelty potential: Low — Quality assessment is well-studied

Feasibility: High — Can build on existing quality assessment methods

Experimental approach:

- Define inspection-specific quality metrics (framing, occlusion, surface visibility, resolution)
- Integrate metrics into quality assessment and viewpoint planning
- Validate on physical robot platform

Publication potential: Medium — Incremental quality metric development is harder to publish

Gap 6.3: Humanoid Platform Validation

Evidence: 27 of 30 papers do not evaluate humanoid robots; only 3 papers address humanoid-specific challenges 3 , 15 , 16

Scientific significance: Medium — Humanoid quality control is important but not fundamentally novel

Novelty potential: Medium — Humanoid-specific quality control is underexplored

Feasibility: Medium — Requires humanoid platform access

Experimental approach:

- Evaluate quality control methods on humanoid platform
- Quantify effects of whole-body motion and camera placement
- Validate on physical humanoid platform

Publication potential: Medium — Humanoid robotics is high-impact, but quality control is less central

Gap ranking: 6.1 > 6.2 > 6.3 (based on novelty, feasibility, publication potential)

B.1.8 RVI-07: Simulation-to-Reality Transfer — Gap Analysis

Gap 7.1: Visibility Model Validation

Evidence: 28 of 30 papers focus on visual domain adaptation or manipulation success; only 2 papers address visibility or coverage validation 1 , 2

Scientific significance: Very high — Visibility model validation bridges simulation and reality for geometric reasoning, not just visual appearance

Novelty potential: Very high — Visibility model validation is underexplored

Feasibility: Medium — Requires physical robot experiments and independent measurement systems

Experimental approach:

- Validate simulated visibility models against physical measurements (laser trackers, photogrammetry)
- Quantify sim-to-real gap in coverage metrics, occlusion detection, surface visibility
- Calibrate simulation parameters to match physical observations

Publication potential: Very high — Sim-to-real transfer is high-impact; visibility validation is novel

Gap 7.2: Geometric Fidelity

Evidence: 24 of 30 papers focus on visual domain adaptation; only 6 papers address geometric or kinematic calibration 4 , 5 , 6 , 11 , 17 , 23

Scientific significance: High — Geometric fidelity is critical for inspection but underexplored in sim-to-real transfer

Novelty potential: Medium — Geometric calibration is studied but not in sim-to-real context

Feasibility: Medium — Requires physical robot experiments and calibration

Experimental approach:

- Calibrate geometric and kinematic parameters (camera intrinsics, extrinsics, robot kinematics)
- Validate simulated geometric reasoning against physical measurements
- Quantify sim-to-real gap in geometric accuracy

Publication potential: High — Geometric fidelity is important for inspection

Gap 7.3: Humanoid Platform Validation

Evidence: 28 of 30 papers evaluate manipulators or wheeled robots; only 2 papers address humanoid-specific challenges 9 , 14

Scientific significance: High — Humanoid sim-to-real transfer is challenging due to bipedal stability and whole-body kinematics

Novelty potential: High — Humanoid-specific sim-to-real transfer is underexplored

Feasibility: Medium — Requires humanoid platform access

Experimental approach:

- Validate sim-to-real transfer on humanoid platform (Unitree G1)
- Quantify sim-to-real gap in locomotion, manipulation, inspection tasks
- Compare against manipulator and wheeled robot baselines

Publication potential: High — Humanoid robotics is high-impact

Gap ranking: 7.1 > 7.3 > 7.2 (based on novelty, feasibility, publication potential)

B.1.9 RVI-08: End-to-End Autonomous Inspection — Gap Analysis

Gap 8.1: Quantitative Autonomous vs. Manual Comparison

Evidence: 26 of 30 papers lack rigorous experimental comparison; only 4 papers compare autonomous vs. manual inspection [1, 7, 22, 26]

Scientific significance: High — Quantitative comparison provides empirical evidence of autonomous inspection performance

Novelty potential: Medium — Comparison studies are valuable but not fundamentally novel

Feasibility: Low — Requires fully functional system and controlled experimental design

Experimental approach:

- Design controlled experimental comparison with standardized tasks and metrics
- Recruit human participants for manual inspection baseline
- Quantify differences in coverage, quality, time, repeatability
- Conduct statistical analysis (ANOVA, post-hoc comparisons)

Publication potential: Medium — Comparison studies are valuable but harder to publish in top venues

Gap 8.2: Documentation Quality Evaluation

Evidence: 25 of 30 papers emphasize defect detection accuracy; only 5 papers address documentation quality [1, 2, 7, 21, 26]

Scientific significance: Medium — Documentation quality is important for inspection but not fundamentally novel

Novelty potential: Low — Quality assessment is well-studied

Feasibility: Medium — Can build on existing quality assessment methods

Experimental approach:

- Define documentation quality metrics (coverage, resolution, framing, illumination)
- Evaluate autonomous inspection against manual baseline
- Validate on physical robot platform

Publication potential: Low — Incremental quality evaluation is harder to publish

Gap 8.3: Humanoid Platform Evaluation

Evidence: 28 of 30 papers do not evaluate humanoid robots; only 2 papers address humanoid-specific challenges [1, 20]

Scientific significance: Medium — Humanoid end-to-end inspection is important but not fundamentally novel

Novelty potential: Medium — Humanoid-specific end-to-end inspection is underexplored

Feasibility: Low — Requires fully functional humanoid inspection system

Experimental approach:

- Develop end-to-end humanoid inspection system
- Compare against manipulator and wheeled robot baselines
- Validate on physical humanoid platform

Publication potential: Medium — Humanoid robotics is high-impact, but system integration is less novel

Gap ranking: 8.1 > 8.3 > 8.2 (based on novelty, feasibility, publication potential)

B.1.10 Gap Summary and Prioritization

Tier 1: High-Priority Gaps (Very High Scientific Significance and Publication Potential)

1. Gap 1.1: Inspection-specific benchmarking protocols
2. Gap 3.1: Humanoid-specific viewpoint and motion planning
3. Gap 6.1: Integrated quality-aware planning
4. Gap 7.1: Visibility model validation

Tier 2: Medium-Priority Gaps (High Scientific Significance and Publication Potential)

5. Gap 1.2: Humanoid platform validation in benchmarking
6. Gap 3.3: Quality integration in viewpoint planning
7. Gap 4.1: Quality-aware NBV planning
8. Gap 4.2: Humanoid platform evaluation in active coverage
9. Gap 7.3: Humanoid platform validation in sim-to-real transfer

Tier 3: Lower-Priority Gaps (Medium Scientific Significance and Publication Potential)

10. Gap 2.1: Multi-instance identification in clutter
11. Gap 5.1: Acquisition geometry repeatability
12. Gap 8.1: Quantitative autonomous vs. manual comparison

B.2 Publication Strategy

B.2.1 Publication Readiness Assessment Framework

Each research direction is assessed for independent, defensible scientific contribution using the following criteria:

1. Scientific novelty: Does the work address a fundamentally new problem or offer a novel solution?
2. Experimental rigor: Does the work include rigorous experimental validation with statistical analysis?
3. Reproducibility: Are methods, datasets, and code publicly available for independent replication?
4. Impact: Does the work advance the state of the art and enable future research?
5. Venue alignment: Is the work aligned with the scope and standards of target venues?

Publication readiness is categorized as:

- High: Ready for submission to top-tier venues (IEEE TRO, IJRR, ICRA, IROS, RSS)
- Medium: Ready for submission to mid-tier venues (IEEE RA-L, Autonomous Robots, specialized journals)

- Low: Requires substantial additional work before publication readiness

B.2.2 RVI-01: Reproducible Benchmarking — Publication Strategy

Publication readiness: High

Target venues:

- Journals: IEEE Transactions on Robotics, International Journal of Robotics Research, Robotics and Autonomous Systems
- Conferences: ICRA, IROS, RSS, CoRL

Rationale: Benchmarking papers attract broad interest and citations; top venues prioritize reproducibility and standardization

Recommended publication sequence:

1. Initial paper: Benchmark framework and evaluation protocol (ICRA/IROS)
2. Follow-up paper: Extended benchmark with additional platforms and tasks (IEEE TRO)
3. Dataset paper: Public release of benchmark datasets and code (IJRR or specialized data journal)

Key contributions:

- Standardized evaluation protocol for autonomous visual inspection
- Controlled environmental variations (lighting, object placement, occlusion)
- Multi-platform validation (manipulator, mobile robot, humanoid)
- Open-source implementation and datasets

Publication timeline: 12-18 months (6 months development, 6 months evaluation, 6 months writing/revision)

B.2.3 RVI-03: Viewpoint and Motion Planning — Publication Strategy

Publication readiness: High

Target venues:

- Journals: IEEE Transactions on Robotics, Autonomous Robots, IEEE Robotics and Automation Letters
- Conferences: ICRA, IROS, RSS, WAFR

Rationale: Planning research is central to robotics; humanoid-specific methods offer novelty

Recommended publication sequence:

1. Initial paper: Humanoid-specific viewpoint planning algorithm (ICRA/IROS)
2. Follow-up paper: Physical validation on Unitree G1 (IEEE TRO)
3. Extended paper: Comparative analysis across platforms (Autonomous Robots)

Key contributions:

- Humanoid-specific viewpoint planning incorporating whole-body kinematics and bipedal stability constraints
- Joint optimization of viewpoint selection and motion planning
- Physical validation on Unitree G1 humanoid platform

- Comparative analysis against manipulator and wheeled robot baselines

Publication timeline: 18-24 months (9 months development, 6 months physical validation, 9 months writing/revision)

B.2.4 RVI-06: Acquisition Quality Control — Publication Strategy

Publication readiness: High

Target venues:

- Journals: IEEE Transactions on Robotics, Computer Vision and Image Understanding, IEEE Transactions on Image Processing
- Conferences: ICRA, IROS, CVPR, ICCV

Rationale: Quality control bridges robotics and computer vision; integrated quality-aware planning is novel

Recommended publication sequence:

1. Initial paper: Quality-aware NBV planning algorithm (ICRA/IROS or CVPR/ICCV)
2. Follow-up paper: Physical validation and comparison against reactive quality control (IEEE TRO)
3. Extended paper: Integration with viewpoint planning and coverage optimization (CVIU)

Key contributions:

- Integrated quality-aware planning combining exposure control, quality assessment, and viewpoint selection
- Proactive quality-aware viewpoint planning vs. reactive quality control
- Physical validation on robot platform
- Inspection-specific quality metrics (framing, occlusion, surface visibility)

Publication timeline: 18-24 months (9 months development, 6 months physical validation, 9 months writing/revision)

B.2.5 RVI-04: Active Visual Coverage — Publication Strategy

Publication readiness: Medium

Target venues:

- Journals: IEEE Transactions on Robotics, Autonomous Robots, IEEE Robotics and Automation Letters
- Conferences: ICRA, IROS, RSS

Rationale: Active perception is a core robotics topic; quality-aware NBV is novel

Recommended publication sequence:

1. Initial paper: Quality-aware NBV planning algorithm (ICRA/IROS)
2. Follow-up paper: Physical validation and comparison against coverage-first NBV (IEEE RA-L or Autonomous Robots)

Key contributions:

- Quality-aware NBV planning integrating image quality assessment into viewpoint selection
- Comparison against coverage-first NBV with post-hoc quality filtering
- Physical validation on robot platform

Publication timeline: 15-21 months (6 months development, 6 months physical validation, 9 months writing/revision)

B.2.6 RVI-07: Simulation-to-Reality Transfer — Publication Strategy

Publication readiness: Medium

Target venues:

- Journals: IEEE Transactions on Robotics, International Journal of Robotics Research, IEEE Robotics and Automation Letters
- Conferences: ICRA, IROS, RSS, CoRL

Rationale: Sim-to-real transfer is high-impact; visibility model validation is novel

Recommended publication sequence:

1. Initial paper: Visibility model validation methodology (ICRA/IROS or CoRL)
2. Follow-up paper: Physical validation and calibration (IEEE TRO or IJRR)

Key contributions:

- Systematic validation of simulated visibility and coverage metrics against physical measurements
- Calibration methodology for visibility models
- Quantification of sim-to-real gap in geometric reasoning

Publication timeline: 18-24 months (9 months development, 6 months physical validation, 9 months writing/revision)

B.2.7 RVI-02: Target Identification and Localization — Publication Strategy

Publication readiness: Medium

Target venues:

- Journals: IEEE Robotics and Automation Letters, Robotics and Autonomous Systems
- Conferences: ICRA, IROS

Rationale: Object detection is well-studied; incremental improvements are harder to publish in top venues

Recommended publication sequence:

1. Initial paper: Multi-instance identification in cluttered inspection scenarios (ICRA/IROS or IEEE RA-L)

Key contributions:

- Multi-instance identification strategies for cluttered inspection scenarios
- Evaluation on standardized datasets (YCB, COCO)

Publication timeline: 12-15 months (6 months development, 3 months evaluation, 6 months writing/revision)

B.2.8 RVI-05: Repeatable Visual Acquisition — Publication Strategy

Publication readiness: Low

Target venues:

- Journals: IEEE Transactions on Automation Science and Engineering, Measurement Science and Technology
- Conferences: ICRA, IROS, CASE

Rationale: Repeatability is important for measurement applications; less central to robotics research

Recommended publication sequence:

1. Initial paper: Acquisition geometry repeatability methodology (CASE or Measurement Science and Technology)

Key contributions:

- Acquisition geometry repeatability metrics (camera pose, framing, resolution, illumination)
- Validation against independent ground truth measurements

Publication timeline: 18-24 months (9 months development, 6 months physical validation, 9 months writing/revision)

Note: Requires substantial experimental validation before publication readiness

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B.2.9 RVI-08: End-to-End Autonomous Inspection — Publication Strategy

Publication readiness: Low

Target venues:

- Journals: IEEE Transactions on Automation Science and Engineering, Robotics and Computer-Integrated Manufacturing
- Conferences: ICRA, IROS, CASE

Rationale: System integration is valuable but harder to publish in top venues; field deployment studies are well-suited to application-oriented journals

Recommended publication sequence:

1. Initial paper: End-to-end humanoid inspection system (CASE or IEEE TASE)
2. Follow-up paper: Quantitative comparison against manual inspection (Robotics and Computer-Integrated Manufacturing)

Key contributions:

- End-to-end humanoid inspection system integrating all components
- Quantitative comparison against manual inspection baseline
- Field deployment validation

Publication timeline: 24-36 months (12 months development, 6 months physical validation, 12 months writing/revision)

Note: Requires fully functional system before publication readiness

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B.2.10 Publication Strategy Summary

High publication potential (ready for top-tier venues):

1. RVI-01: Reproducible Benchmarking
2. RVI-03: Viewpoint and Motion Planning
3. RVI-06: Acquisition Quality Control

Medium publication potential (ready for mid-tier venues or specialized journals): 4. RVI-04: Active Visual Coverage 5. RVI-07: Simulation-to-Reality Transfer 6. RVI-02: Target Identification and Localization

Lower publication potential (requires substantial additional work): 7. RVI-05: Repeatable Visual Acquisition 8. RVI-08: End-to-End Autonomous Inspection

Recommended publication sequence:

1. Year 1: RVI-01 (benchmarking framework), RVI-03 (humanoid planning algorithm)
2. Year 2: RVI-06 (quality-aware planning), RVI-04 (quality-aware NBV)
3. Year 3: RVI-07 (visibility validation), RVI-02 (multi-instance identification)
4. Year 4: RVI-05 (acquisition repeatability), RVI-08 (end-to-end system)

This sequence prioritizes high-impact, high-novelty contributions while building toward integrated system validation.

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